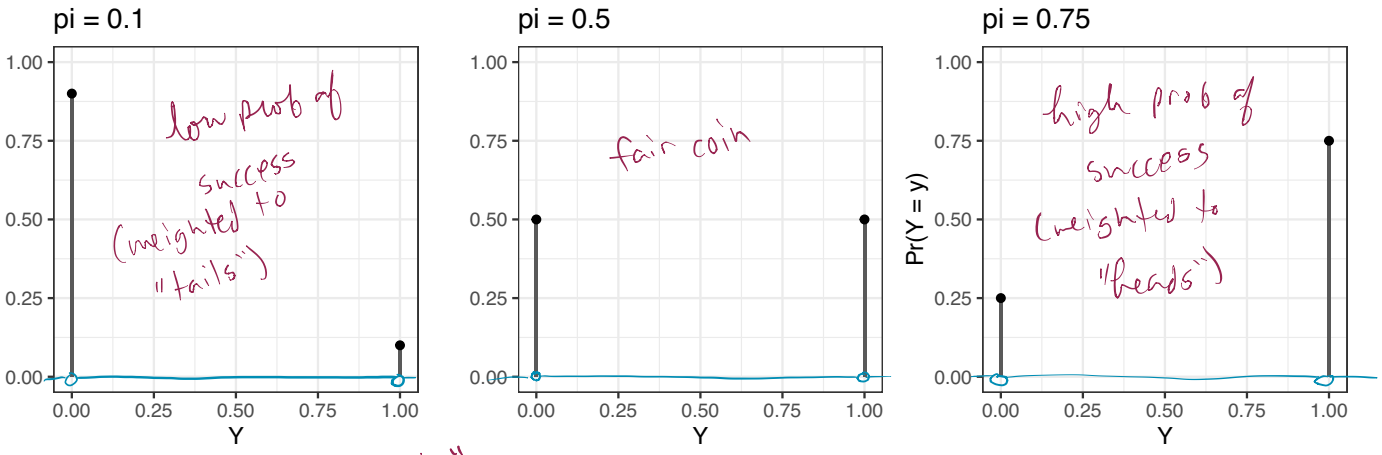
success = heads in weighted coin



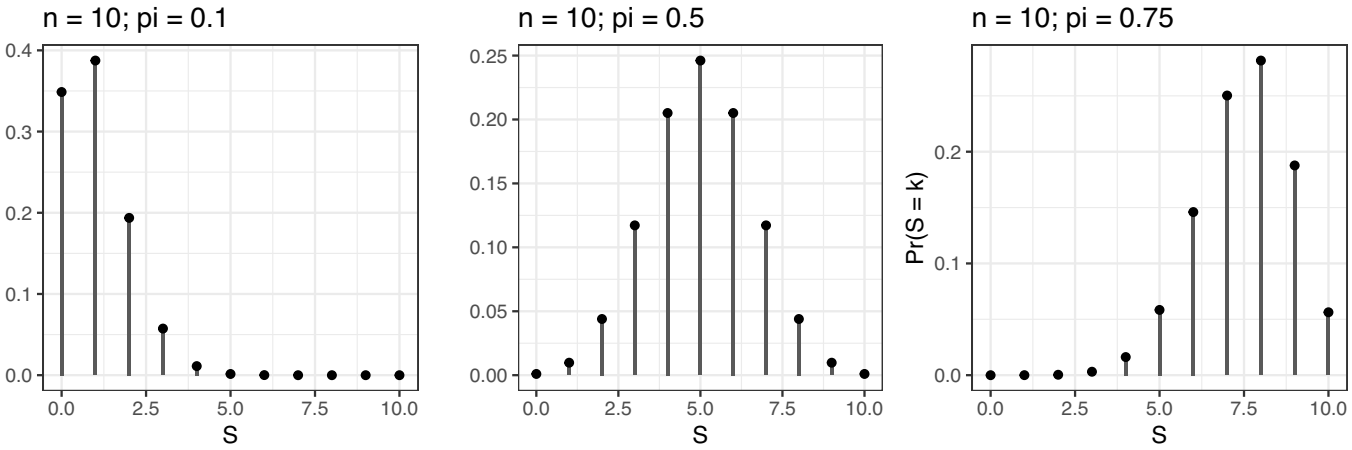

n independent "coin flips"
$Y_i \sim$ Bernoulli($\pi$)

## Binomial distribution

Let $\{Y_1, Y_2, ..., Y_n\}$ be RS from population of binary response variables with probability of yes equal to $\pi$

- $S = Y_1 + Y_2 + ... + Y_n$ is the total number Y = "yes" (1)
- $S \sim Binomial(n, \pi)$
- $Pr[S = k] = \binom{n}{k}\pi^k(1-\pi)^{n-k}$ for $S \in \{0, 1, 2, 3, ..., n\}$ — "n choose k" $\binom{n}{k} = \frac{n!}{k!(n-k)!}$
- $Mean\{S\} = E[S] = n\pi$ "expected yes's" $n(1-\pi)$ = "expected nos"
- $Var\{S\} = E[(S - n\pi)^2] = n\pi(1-\pi)$
- When # expected "yes" $n\pi$ and # expected "no" $n\pi(1-\pi)$ are moderately large, binomial probabilities are approximated by a normal distribution!

## Visually

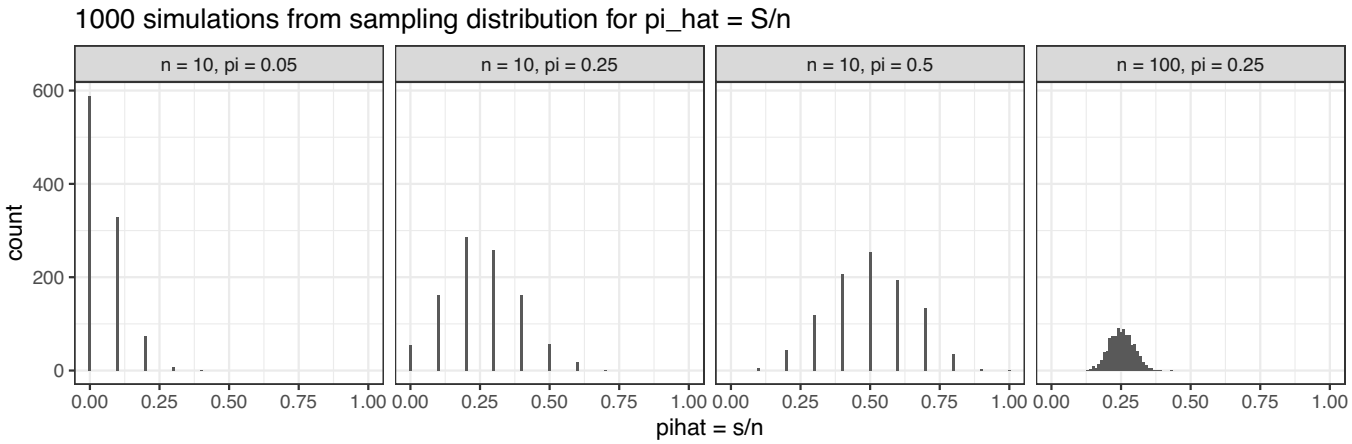



Observe RS of binary response data $\{Y_1, Y_2, ..., Y_n\}$, and want to estimate $\pi$

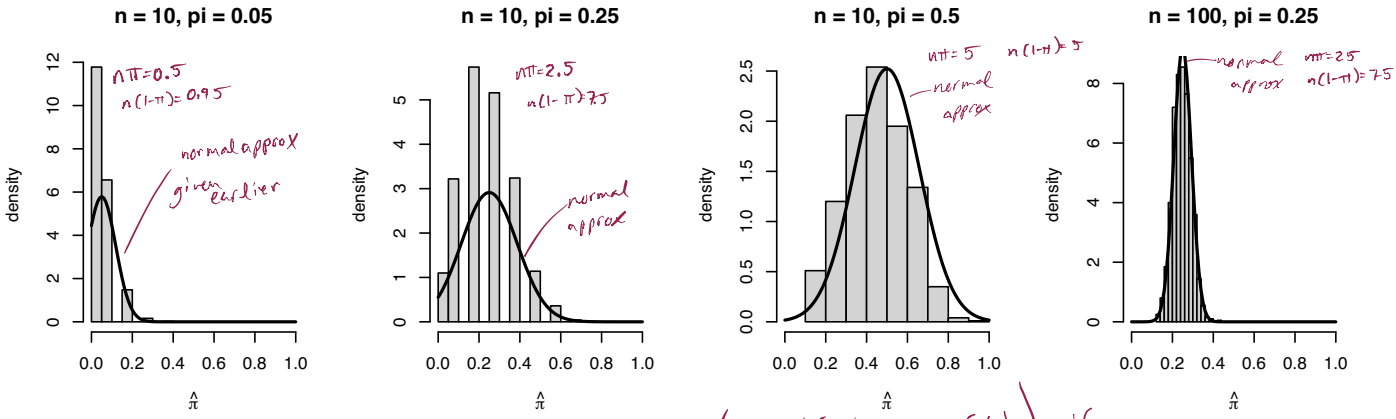
- Point estimate: $\hat{\pi} = S/n$
- $\text{Mean}\{\hat{\pi}\} = \pi$
- $\text{Var}\{\hat{\pi}\} = \pi(1-\pi)/n$
- If $n$ is large enough, the sampling distribution of $\hat{\pi}$ is approximately normal!
  - Use $n\pi_0$ and $n(1-\pi_0)$ to assess if Normality is reasonable

## Visually

$\hat{\pi} = \frac{S}{n}$ observed proportion of success/yes in sample




## Compared to Normal Approximation




$\hat{\pi}$ is best guess at $\pi$ Sampdist of $\hat{\pi}$ is $\hat{\pi}$ is $N(\text{mean}(S/n), \text{var}(S/n))$ if $n\pi$; $n(1-\pi)$ is large enough

Observe RS $\{Y_1, Y_2, ..., Y_{n_1}\}$ from group 1 and RS from $\{Y_1, Y_2, ..., Y_{n_2}\}$ group 2 and want to estimate $\pi_1 - \pi_2$

- Point estimate: $\hat{\pi}_1 - \hat{\pi}_2 = S_1/n_1 - S_2/n_2$
- Mean$\{\hat{\pi}_1 - \hat{\pi}_2\} = \pi_1 - \pi_2$
- Var$\{\hat{\pi}_1 - \hat{\pi}_2\} = \frac{\pi_1(1-\pi_1)}{n_1} - \frac{\pi_2(1-\pi_2)}{n_2}$
- If $n_1$ and $n_2$ are large, the sampling distribution of $\hat{\pi}_1 - \hat{\pi}_2$ is approximately normal!
  - Use smaller of the sample sizes and a combined prob of yes to check that expected # of yes and expected number of no are "large enough"

## Ch 18 has much more...

- Formulas for CIs and hypothesis tests for one mean and comparing two means are provided
- Analogous to X11 methods for inference for one mean or a difference in two groups means
- For comparing proportions we use Z-tests (Normal) rather than T-tests. Why?

b/c only estimating $\pi$ and $M\{\hat{\pi}\} = \pi$ — extra uncertainty

$Y_i$ is binary, $Y_i \sim \text{Bern}(\pi)$, $\text{var}\{\hat{\pi}\} = \pi(1-\pi)/n$, 1 param vs $Y_i \sim N(\mu, \sigma^2)$, $M\{\bar{Y}\} = \mu$, 2 params, $\text{var}\{\bar{Y}\} = \sigma^2/n$

## Differences can be misleading

What values of $\pi_1$ and $\pi_2$ make the following true?

$$\pi_1 - \pi_2 = 0.05$$

⚠ Practical implication of $\pi_1 - \pi_2$ is non-constant!

Depends on the actual values of $\pi_1$ and $\pi_2$.

- When $\pi_1$ and $\pi_2$ are close to 0 or 1, practical implications tend to be much larger than when the parameter values are close to 0.5

## So what do we do?

- We can compare odds across groups instead of differences in probabilities

### Properties of odds

$\omega = \frac{\pi}{1-\pi}$ "probability of success/yes in pop divided by probability of failure/no in pop"

1. $\omega \in [0, \infty)$
2. $\pi = 0.5 \implies \omega = 1$
3. If odds of "yes" $= \omega$, odds of "no" $= \frac{1}{\omega}$
4. If odds of "yes" $= \omega$, probability of "yes" $\pi = \frac{\omega}{1+\omega}$

### Practice interpreting odds

$\omega_1 = 5 \implies$ in group 1, there are 5 yes for every no

"odds of yes in pop1/group1"

5:1 → yes:no

$\omega_1 = 0.5 \implies \frac{1}{\omega_1}$ = "odds of no" $\frac{1}{\omega_1} = \frac{1}{0.5} = 2$

"odds of yes in pop1 is 0.5" → in group 1 there are 2 nos for every yes

yes:no (1:2) odds

$\omega_1 = 0.8 \implies$ 0.8 yes of every no

"odds of yes in pop1 is 0.8" in group 1 there are 0.8(10)=8 yeses for every 10(10)=10 nos

### Estimating $\pi$ and $\omega$

| | Y = *yes* | Y = *no* | Totals |
|---|---|---|---|
| X = Group 1 | $n_{1,yes}$ | $n_{1,no}$ | $n_1 = R_1$ |
| X = Group 2 | $n_{2,yes}$ | $n_{2,no}$ | $n_2 = R_2$ |
| | $n_{yes} = C_1$ | $n_{no} = C_2$ | $n = T$ |

$\hat{\pi}_2$ estimates $\pi_2$ $\hat{\pi}_2 = \frac{n_{2,yes}}{n_2}$

$\hat{\pi}_1$ **estimates** $\pi_1$ $\hat{\pi}_1 = \frac{n_{1,yes}}{n_1} = \frac{\#\text{ of yes in grp1}}{\#\text{ in grp1}}$

$\hat{\omega}_1$ **estimates** $\omega_1$ $\hat{\omega}_1 = \frac{\hat{\pi}_1}{1-\hat{\pi}_1}$

$\hat{\omega}_2$ estimates $\omega_2$ $\hat{\omega}_2 = \frac{\hat{\pi}_2}{1-\hat{\pi}_2}$

### Odds Ratio

"phi"

$\phi = \omega_1/\omega_2$ is typically used to compare probabilities across groups

1. $\phi$ tends to remain more nearly constant (than $\pi_1 - \pi_2$) over levels of confounding variables
2. $\phi$ is only valid measure for *retrospective studies*
3. $\phi$ extends to logistic regression! (Ch 20/21)

$\hat{\phi}$ **estimates** $\phi$ $\hat{\phi} = \widehat{\omega}_1 / \widehat{\omega}_2 = \hat{\phi}$

"$\phi$ odds of "yes" in group 1 relative to group 2"

## Stat theory in a nutshell

If $\widehat{\omega_1}$ and $\widehat{\omega_2}$ are computed from independent samples, then the *sampling distribution of the log-odds* $log(\widehat{\omega_1}/\widehat{\omega_2})$...

1. Has approximate center equal to $log(\omega_1/\omega_2)$ ①
2. Has approximate variance equal to $\frac{1}{n_1\pi_1(1-\pi_1)+n_2\pi_2(1-\pi_2)}$ ②
3. If both $n_1$ and $n_2$ are large, $log(\widehat{\omega_1}/\widehat{\omega_2}) \dot{\sim} N(mean(1), var(2))$ (1 = ①, 2 = ②)

seen log-odds before with seaweed grazers

↳ now we will have the tools to do this correctly

## Statistical inference for large $n_1$, $n_2$ (CIs)

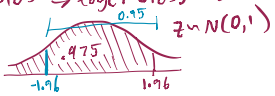
$100(1-\alpha)\%$ CI for $\log(\phi)$: estimated SD of distribution for $\log(\hat{\phi})$

$$\log(\hat{\phi}) \pm z^*_{1-\alpha/2} \, SE(\log(\hat{\phi}))$$

$$SE(\log(\hat{\phi})) = \sqrt{\frac{1}{n_1\hat{\pi}_1(1-\hat{\pi}_1)} + \frac{1}{n_2\hat{\pi}_2(1-\hat{\pi}_2)}}$$

Ex: 95% CI for $\log(\phi)$; $\alpha = 0.05 \Rightarrow \log(1-0.05) = .95$

$z^*_{1-\frac{0.05}{2}} = 1.96$




## Statistical inference for large $n_1$, $n_2$ (tests)

$H_0: \phi = 1 \Rightarrow \log(\phi) = 0$

$H_a: \phi > 1 \Rightarrow \log(\phi) > 0$

$\phi < 1 \Rightarrow \log(\phi) < 0$

$\phi = 1 \quad \neq 0$

T.S. $z = \frac{\log(\hat{\phi})}{SE(\log(\hat{\phi}))} \therefore z \sim N(0,1)$ under $H_0$

p-value

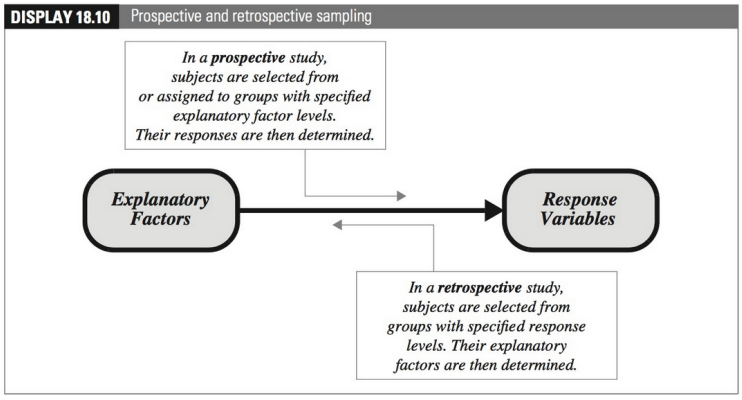
* $SE(\log(\hat{\phi}))$ is calculated with a common $\hat{\pi}_C \Rightarrow$ see book
* slightly different formula

## Back to smoking and lung cancer

Interested in lung cancer incidence (yes/no) in two groups smokers (1) and non-smokers (2).

- Can we use a randomized design to assign smoking to participants?
- Can we identify groups of smokers and non-smokers within the same population and see if they develop lung cancer?
- More practical to sample participants with lung cancer and those without from the same population and ask them if they are a smoker.

## Retrospective vs. prospective studies



## $\phi$ for retrospective studies

1954 retrospective study

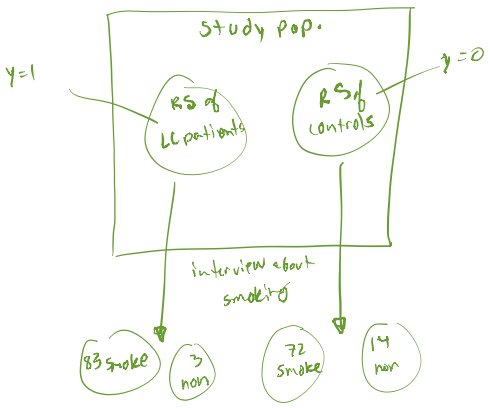
**DISPLAY 18.11** Illustration that the odds ratio is the same regardless of which factor is thought of as the response, with a hypothetical population

| | Lung cancer | No cancer |
|---|---|---|
| Smokers | 1,000 | 2,000,000 |
| Nonsmokers | 4,000 | 16,000,000 |

$$\frac{\text{Odds of cancer among smokers}}{\text{Odds of cancer among nonsmokers}} = \frac{1{,}000/2{,}000{,}000}{4{,}000/16{,}000{,}000} = 2$$

$$\frac{\text{Odds that a cancer victim was a smoker}}{\text{Odds that a person without cancer was a smoker}} = \frac{1{,}000/4{,}000}{2{,}000{,}000/16{,}000{,}000} = 2$$

only valid parameter for retrospective studies




## Concept diagram for case study

## Create 95% CI for log($\phi$)

retrospective study

| | Cancer | Control | Totals |
|---|---|---|---|
| Smokers | 83 | 72 | 155 |
| Nonsmokers | 3 | 14 | 17 |
| Totals | 86 | 86 | 172 |

## Point Estimates

```
# sample prob cancer in smoking group
(phat1 <- 83/155)
```

```
[1] 0.5354839
```

```
# sample odds
(omhat1 <- phat1/(1-phat1))
```

```
[1] 1.152778
```

```
# sample prob cancer in non-smoking group
(phat2 <- 3/17)
```

```
[1] 0.1764706
```

```
# sample odds
(omhat2 <- phat2/(1-phat2))
```

```
[1] 0.2142857
```

```
# sample odds ratio
(phi_hat <- omhat1/omhat2)
```

```
[1] 5.37963
```

## 95% CI for $log(\phi)$

```
# 95% CI for log(phi): log_phihat +/- 1.96*SE(log(phihat))
lphi_hat <- log(phi_hat)
lphi_hat
```

```
[1] 1.68262
```

```
se_lphi_hat <- sqrt(1/(155*phat1*(1-phat1)) + 1/(17*phat2*(1-phat2)))
se_lphi_hat
```

```
[1] 0.6562766
```